

FR350SDJ Rear Helper Spring Kit Installation Manual

Model: FR350SDJ (Rear Suspension Package)

Please read these instructions carefully before commencing installation. Ensure all standard safety protocols are observed when working beneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Hollow Rubber Spring	2
3	Rubber Spacer (4" O.D. x 1/2" Thick with P Bushing)	2
4	Hex Cap Screw (HCS) M10 x 50 x 1.5 FT Gr 8.8 Z	2
5	Shim (1-3/8" L x 1/8" W x 1/32" T x 5/8" Slot)	4

2. Installation Instructions

- 1. Raise Vehicle:** Raise the vehicle using an appropriate hoist or jack, allowing the axle to hang freely. Securely support the chassis before proceeding with any work.
- 2. Remove Bump Stop:** Remove the existing original equipment manufacturer (OEM) bump stop assembly. Retain the original mounting hardware for later use.
- 3. Assemble Helper Spring Unit:** Assemble one hollow rubber spring, one rubber spacer, and one weldment bracket. Secure these components together using one M10 x 50 hex cap screw. Repeat this assembly process for the opposite side of the vehicle.
- 4. Mount Assembly:** Mount each assembled helper spring unit to the vehicle frame using the existing mounting holes. Utilize the retained OEM fasteners and the provided shims as necessary to achieve proper alignment and fitment.
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- Verify that brake lines or cables do not interfere with the helper spring assembly, especially during full suspension travel.
- If the vehicle's original equipment includes a peg-style casting, the helper spring is designed to make direct contact with this component. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** Rear helper spring kits for light-duty trucks and vans should maintain a 1/2" to 1-1/2" gap between the rubber spring and the axle contact point when the vehicle is unladen.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or eliminate this initial clearance gap.
- **Heavy-Duty Applications:** For medium and heavy-duty trucks, trailers, and motorhomes, the helper spring assembly is designed for constant contact with no initial gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended Gross Axle Weight Rating (GAWR).
- Inspect for potential interference between the hollow rubber spring and other vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.